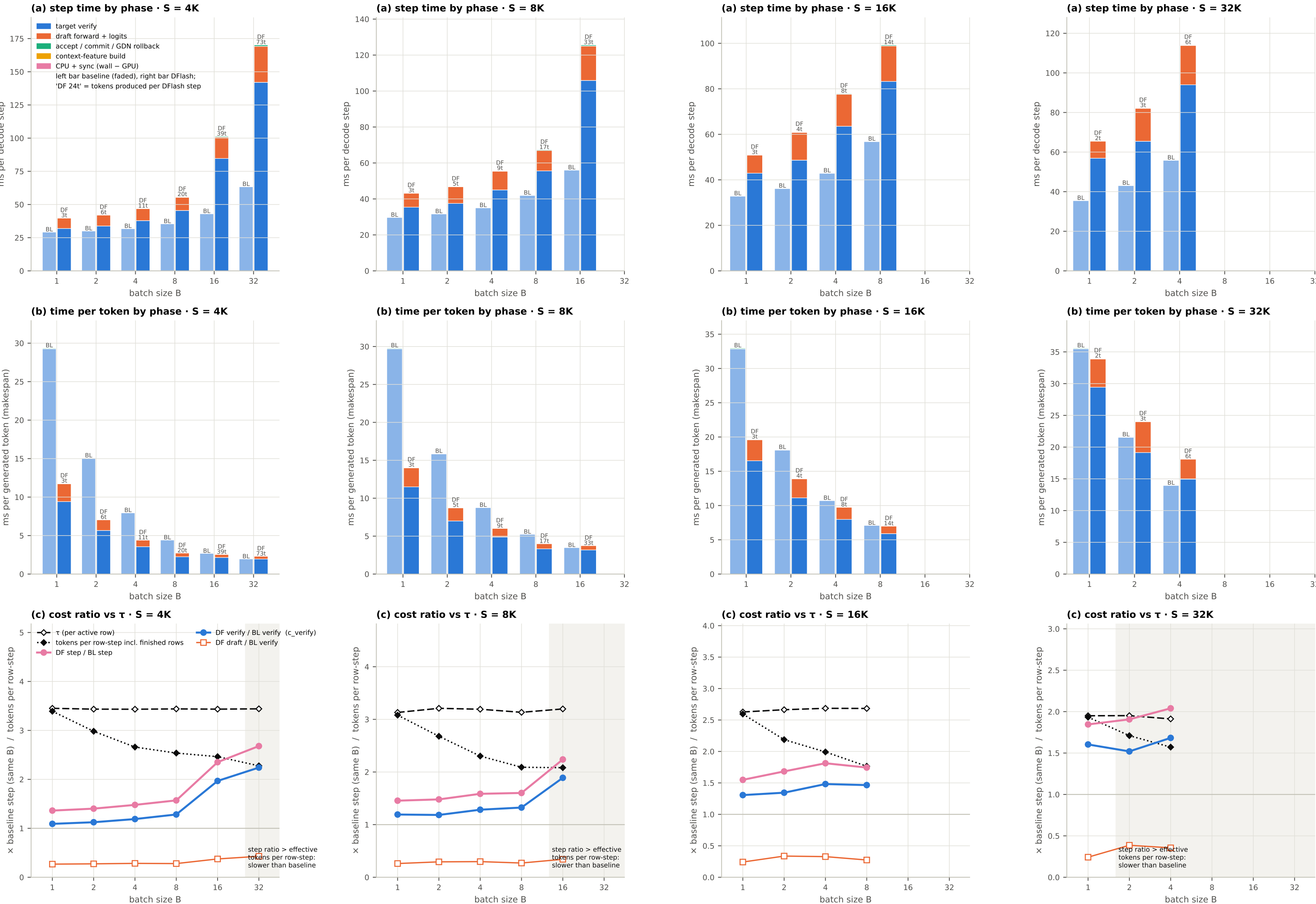


Qwen3-8B — where a decode step's time goes as B grows



GPU phases are CUDA-event means per step; CPU + sync is wall - Σ GPU (can be slightly negative, drawn as 0). (c) Makespan speedup = (tokens per row-step incl. finished rows) / (DF step / BL step) exactly — the magenta line crossing the dotted line is the break-even. The gap between the dashed τ and the dotted line is the static-batch tail (finished rows still verified). c_verify near 1 means 15 extra tokens per row are nearly free (memory-bound); it rises once the verify is compute-bound.